

Research Article



Efficacy of buccal piroxicam infiltration and inferior alveolar nerve block in patients with irreversible pulpitis: a prospective, double-blind, randomized clinical trial

Saurav Paul , Sridevi Nandamuri , Aakrati Raina ,* Mukta Bansal

Department of Conservative Dentistry and Endodontics, Rama Dental College, Hospital and Research Centre, Kanpur, Uttar Pradesh, India



Received: Mar 27, 2020

Revised: Jun 24, 2020

Accepted: Sep 2, 2020

Paul S, Nandamuri S, Raina A, Bansal M

*Correspondence to

Aakrati Raina, MDS, BDS

Postgraduate Student, Department of Conservative Dentistry and Endodontics, Rama Dental College, Hospital and Research Centre, Kanpur, Uttar Pradesh 208024, India.
E-mail: aakratiraina@gmail.com

Copyright © 2021. The Korean Academy of Conservative Dentistry

This is an Open Access article distributed under the terms of the Creative Commons Attribution Non-Commercial License (<https://creativecommons.org/licenses/by-nc/4.0/>) which permits unrestricted non-commercial use, distribution, and reproduction in any medium, provided the original work is properly cited.

Conflict of Interest

No potential conflict of interest relevant to this article was reported.

Author Contributions

Conceptualization: Paul S, Nandamuri S;
Data curation: Paul S, Nandamuri S, Bansal M; Formal analysis: Paul S, Raina A; Funding acquisition: Paul S, Nandamuri S, Raina A, Bansal M; Investigation: Paul S, Raina A, Bansal M; Methodology: Paul S, Nandamuri S, Raina A, Bansal M; Project administration: Paul S, Nandamuri S, Raina A, Bansal M; Resources: Paul S, Nandamuri S, Raina A; Software: Paul

Keywords: Anesthesia; Buccal infiltration; Inferior alveolar nerve block; Irreversible pulpitis; Lignocaine; Piroxicam

INTRODUCTION

Irreversible pulpitis is a pulpal pathosis characterized by an exaggerated painful thermal response, particularly to cold stimuli, wherein the inflamed pulp is incapable of healing and returning to a normal state. Pain can either be intensified by a stimulus or be spontaneous [1]. The principal treatment for this condition is root canal therapy [2].

S, Bansal M; Supervision: Raina A; Validation: Paul S, Nandamuri S; Visualization: Paul S, Raina A, Bansal M; Writing - original draft: Paul S, Nandamuri S, Raina A, Bansal M; Writing - review & editing: Paul S, Nandamuri S.

ORCID iDs

Saurav Paul 
<https://orcid.org/0000-0003-2941-6846>
Sridevi Nandamuri 
<https://orcid.org/0000-0003-2379-8361>
Aakrati Raina 
<https://orcid.org/0000-0001-8066-2407>
Mukta Bansal 
<https://orcid.org/0000-0001-5898-4020>

Complete pulpal anesthesia, particularly inferior alveolar nerve block (IANB), may be difficult to attain in mandibular molars with irreversible pulpitis, and the presence of irreversible pulpitis therefore reduces the success rate of IANB [3-6]. Studies have reported that achieving successful anesthesia in inflamed pulp is 8 times more difficult than in normal pulp, with the failure rate of IANB ranging from 44% to 81% [4,6,7]. Explanations for this difficulty include the tachyphylaxis of the anesthetic solution, decreased pH, and stimulation and modifications of the response of nociceptors (including tetrodotoxin-sensitive and capsaicin-sensitive transient receptor potential vanilloid type 1 [TRPV1] nociceptors) in the presence of inflammatory mediators such as prostaglandins (PGs) [4,8].

Nonsteroidal anti-inflammatory drugs (NSAIDs), which deactivate the cyclooxygenase (COX) pathway, have an inhibitory impact on PG production [9]. Thus, it has been hypothesized that in irreversible pulpitis, premedication with NSAIDs helps to achieve anesthesia. Some studies have reported that supplemental infiltration with ketorolac tromethamine and dexamethasone increased the effectiveness of IANB in irreversible pulpitis [4,10].

Piroxicam, an oxycam derivative with pain-relieving and anti-inflammatory effects, is a non-selective COX inhibitor that acts by inhibiting PG synthesis [11,12]. A study reported significant improvement in pain within 1 hour of the initial dose that improved over a 12-hour period and lasted for 24 hours after parenteral administration of piroxicam [13]. To date, no study in the literature has evaluated the efficacy of buccal piroxicam infiltration on the success of IANB. Hence, this prospective, double-blind, randomized trial was conducted to gauge the influence of buccal infiltration with piroxicam on the success of IANB and buccal infiltration using 2% lignocaine with adrenaline (1:80,000) in patients with irreversible pulpitis.

MATERIALS AND METHODS

After receiving approval from the Ethics Committee of the institution and registering the trial with the Clinical Trial Registry-India (CTRI Reg. No.-CTRI/2018/10/016010), the study was commenced. This trial included 56 patients, aged 15 to 65 years, who visited the Department of Endodontics at Rama Dental College, Hospital and Research Centre, Kanpur, India.

The effect size was computed using G*Power 3.1 software (Heinrich-Heine-Universität Düsseldorf, Dusseldorf, Germany) as the least difference expected between groups, using the mean visual analogue scale (VAS) scores of patients from a previous study [4]. The sample size was then calculated using an effect size of 0.82 at $\alpha = 0.05$ and a power of 0.95, resulting in a total sample size for the 2 groups being 84. For a power of 0.8, the sample size would be 52, thus, the calculated sample size for our study was 52 (26 in each group). To control for the possibility of manual errors, we rounded the sample size to 56 (28 in each group). Each patient completed a written informed consent form, as well as an oral and written questionnaire to assess his or her health and preoperative pain. Before commencing the procedure, each patient was taught how to score his or her pain on the 170-mm Heft-Parker VAS (HP-VAS), with 0 indicating no pain, 1-54 indicating slight pain, 55-114 indicating moderate pain, and > 114 indicating extreme pain.

Patients with pain in the mandibular molars (> 54 on the HP-VAS), vital teeth with no distinct periapical pathologies, an exaggerated response to cold testing (Endo-Frost, Coltene Whaledent, Langenau, Germany) with pain lasting > 45 seconds, an initial response

recorded as 0–40 with an electric pulp tester (EPT) (COXO SOCO C-Pulse pulp tester, Foshan COXO Medical Instrument Co. Ltd, Guangdong, China) were included in this study if they were categorized as having either a class I or II medical history (American Society of Anesthesiologists) and could comprehend the consent form and the HP-VAS. Individuals with more than 1 infected tooth; an inability to achieve lip numbness with a single infusion of IANB; known hypersensitivity or allergy to piroxicam or other anti-inflammatory drugs; or a history of peptic ulcers, renal, liver or bleeding disorders were excluded from the study, as were pregnant or lactating patients.

Each patient was handed a packet containing a consent form, questionnaire, a bottle of 2% lignocaine with adrenaline (1:80,000; Lignox 2%, Indoco Remedies, Gandhinagar, Gujarat, India) and 25-gauge syringes (Hi-Tech, Mumbai, India). Normal saline (Pentagon Labs, Dewas, Madhya Pradesh, India) and piroxicam (Dolonex, Pfizer Ltd., Mumbai, India) were loaded into disposable syringes and coded as A and B, respectively.

A standard IANB injection using 1.8 mL of 2% lignocaine with 1:80,000 epinephrine, immediately followed by buccal infiltration of 0.9 mL lignocaine, was administered to every patient. After 5 minutes, randomized allocation ensuring blinding of the subjects was done in 2 groups ($n = 28$) by a skilled postgraduate student who was ignorant of the study design and purpose and was requested to provide supplemental buccal infiltration with syringe A and syringe B for the patients in groups 1 and 2, respectively. Thus, group 1 was the control group (buccal infiltration with normal saline, syringe A), and group 2 was the test group (buccal infiltration with 20 mg/mL of piroxicam, syringe B).

Pulpal anesthesia was evaluated every 5 minutes by using an analogue EPT and assessing lip numbness. An access opening procedure (AOP) was performed 15 minutes post-IANB once the patient showed signs of lip numbness and gave 2 successive negative responses to the EPT. If not, the patient was not included in the study.

The 2 groups were compared in terms of patients' age and sex (**Table 1**). The HP-VAS was used to grade the patient's initial pain score (IPS) and pain during caries removal (CR), AOP, and working length measurement (WLM) (**Table 2**). Successful anesthesia was identified by either the absence of pain or mild/slight pain during CR, AOP, and WLM, with no requirement for a further anesthetic dose. Any additional anesthetic dose, if required due to severe pain, led to the exclusion of that patient from the trial.

mean pain scores during various procedures in the test (piroxicam) and control (saline)

CONCLUSIONS

Within the constraints of the present trial, we conclude that piroxicam buccal infiltration significantly enhances the successful achievement of IANB accompanied with buccal infiltration of lignocaine in individuals with irreversible pulpitis.

ACKNOWLEDGEMENTS

The authors are grateful to every patient involved in this research for providing his/her consent, thereby allowing the completion of the trial.

REFERENCES

1. Handsides RA, Jaramillo DE, Ingle JI. Examination, evaluation, diagnosis and treatment planning. In: Ingle JI, Bakland LK, Baumgartner JC, editors. Ingle's endodontics. 6th ed. Hamilton, ON; BC Decker Inc.; 2008. Chapter 14.
2. Ali SG, Mulay S. Pulpitis: a review. IOSR J Dent Med Sci 2015;14:92-97.
3. Aggarwal V, Singla M, Kabi D. Comparative evaluation of effect of preoperative oral medication of ibuprofen and ketorolac on anesthetic efficacy of inferior alveolar nerve block with lidocaine in patients with irreversible pulpitis: a prospective, double-blind, randomized clinical trial. J Endod 2010;36:375-378.
[PUBMED](#) | [CROSSREF](#)
4. Akhlaghi NM, Hormozi B, Abbott PV, Khalilak Z. Efficacy of ketorolac buccal infiltrations and inferior alveolar nerve blocks in patients with irreversible pulpitis: a prospective, double-blind, randomized clinical trial. J Endod 2016;42:691-695.
[PUBMED](#) | [CROSSREF](#)

5. Rogers BS, Botero TM, McDonald NJ, Gardner RJ, Peters MC. Efficacy of articaine versus lidocaine as a supplemental buccal infiltration in mandibular molars with irreversible pulpitis: a prospective, randomized, double-blind study. *J Endod* 2014;40:753-758.
[PUBMED](#) | [CROSSREF](#)
6. Prasanna N, Subbarao CV, Gutmann JL. The efficacy of pre-operative oral medication of lornoxicam and diclofenac potassium on the success of inferior alveolar nerve block in patients with irreversible pulpitis: a double-blind, randomised controlled clinical trial. *Int Endod J* 2011;44:330-336.
[PUBMED](#) | [CROSSREF](#)
7. Verma PK, Srivastava R, Ramesh KM. Anesthetic efficacy of X-tip intraosseous injection using 2% lidocaine with 1:80,000 epinephrine in patients with irreversible pulpitis after inferior alveolar nerve block: a clinical study. *J Conserv Dent* 2013;16:162-166.
[PUBMED](#) | [CROSSREF](#)
8. Hargreaves KM, Keiser K. Local anesthetic failure in endodontics: mechanisms and management. *Endod Topics* 2002;1:26-39.
[CROSSREF](#)
9. Smith WL, DeWitt DL, Garavito RM. Cyclooxygenases: structural, cellular, and molecular biology. *Annu Rev Biochem* 2000;69:145-182.
[PUBMED](#) | [CROSSREF](#)
10. Aggarwal V, Singla M, Rizvi A, Miglani S. Comparative evaluation of local infiltration of articaine, articaine plus ketorolac, and dexamethasone on anesthetic efficacy of inferior alveolar nerve block with lidocaine in patients with irreversible pulpitis. *J Endod* 2011;37:445-449.
[PUBMED](#) | [CROSSREF](#)
11. Brogden RN, Heel RC, Speight TM, Avery GS. Piroxicam: a review of its pharmacological properties and therapeutic efficacy. *Drugs* 1981;22:165-187.
[PUBMED](#) | [CROSSREF](#)
12. Shanbagh TV, Shenoy S, Nayak V. Autacoids and respiratory system. In: Shanbagh TV, Shenoy S, Nayak V, editors. *Pharmacology for dental students*. 2nd ed. Taramani, Chennai: Elsevier; 2010. Chapter 7.
13. Wiseman RL, Kilgour M. Intramuscular piroxicam, a new dosage form, in the treatment of acute musculoskeletal disorders. *J Int Med Res* 1985;13:255-262.
[PUBMED](#) | [CROSSREF](#)
14. Cohen HP, Cha BY, Spångberg LS. Endodontic anesthesia in mandibular molars: a clinical study. *J Endod* 1993;19:370-373.
[PUBMED](#) | [CROSSREF](#)
15. Parirokh M, Satvati SA, Sharifi R, Rekabi AR, Gorjestani H, Nakhaee N, Abbott PV. Efficacy of combining a buccal infiltration with an inferior alveolar nerve block for mandibular molars with irreversible pulpitis. *Oral Surg Oral Med Oral Pathol Oral Radiol Endod* 2010;109:468-473.
[PUBMED](#) | [CROSSREF](#)
16. Parirokh M, Yosefi MH, Nakhaee N, Abbott PV, Manoochchirif H. The success rate of bupivacaine and lidocaine as anesthetic agents in inferior alveolar nerve block in teeth with irreversible pulpitis without spontaneous pain. *Restor Dent Endod* 2015;40:155-160.
[PUBMED](#) | [CROSSREF](#)
17. Kaviani N, Khademi A, Ebtehaj I, Mohammadi Z. The effect of orally administered ketamine on requirement for anesthetics and postoperative pain in mandibular molar teeth with irreversible pulpitis. *J Oral Sci* 2011;53:461-465.
[PUBMED](#) | [CROSSREF](#)
18. Vadivelu N, Gowda AM, Urman RD, Jolly S, Kodumudi V, Maria M, Taylor R Jr, Pergolizzi JV Jr. Ketorolac tromethamine - routes and clinical implications. *Pain Pract* 2015;15:175-193.
[PUBMED](#) | [CROSSREF](#)
19. Joshi N, Mathew S, George JV, Hegde S, Bhandi S, Madhu KS. Comparative evaluation of the efficacy of two modes of delivery of Piroxicam (Dolonex®) for the management of postendodontic pain: a randomized control trial. *J Conserv Dent* 2016;19:301-305.
[PUBMED](#) | [CROSSREF](#)
20. Atbaei A, Mortazavi N. Prophylactic intraligamentary injection of piroxicam (feldene) for the management of post-endodontic pain in molar teeth with irreversible pulpitis. *Aust Endod J* 2012;38:31-35.
[PUBMED](#) | [CROSSREF](#)
21. Subhan Z, Shami S, Ahmed Rana MJ. Comparison of prophylactic intraligamentary injection of piroxicam and lignocaine for management of postoperative endodontic pain. *Pak Oral Dent J* 2016;36:480-483.
22. Yadav S. Anesthetic success of supplemental infiltration in mandibular molars with irreversible pulpitis: a systematic review. *J Conserv Dent* 2015;18:182-186.
[PUBMED](#) | [CROSSREF](#)

23. Goddard G, Karibe H, McNeill C. Reproducibility of visual analog scale (VAS) pain scores to mechanical pressure. *Cranio* 2004;22:250-256.
[PUBMED](#) | [CROSSREF](#)
24. Averbuch M, Katzper M. Assessment of visual analog versus categorical scale for measurement of osteoarthritis pain. *J Clin Pharmacol* 2004;44:368-372.
[PUBMED](#) | [CROSSREF](#)
25. Jena A, Shashirekha G. Effect of preoperative medications on the efficacy of inferior alveolar nerve block in patients with irreversible pulpitis: a placebo-controlled clinical study. *J Conserv Dent* 2013;16:171-174.
[PUBMED](#) | [CROSSREF](#)